Case Study Assignment

AI Chatbot Security & Cryptography

Background

A private hospital in Karachi implemented an AI-powered chatbot system to improve patient services. The chatbot allows users to:
- Book appointments
- Ask medical-related questions
- Access limited personal health information

The chatbot is available via a mobile application and a web portal.

Problem Scenario

Within a short period after deployment, multiple security issues were reported:
- Sensitive patient information (names, reports, medical conditions) was exposed
- Some users reported that their conversations were being accessed by unknown parties
- Unauthorized individuals were able to log into accounts without proper verification
- Fake requests were submitted using other patients' identities
- The system could not detect whether a message was altered during transmission

Due to these issues, the hospital management is concerned about data privacy, system security, and user trust.

Technical Situation

The development team reveals the following about the system:
- Communication between users and chatbot is not encrypted
- There is no proper authentication mechanism
- The system does not verify the identity of users
- No cryptographic techniques (encryption, hashing, digital signatures) are currently implemented
- AI chatbot processes all incoming requests without validation

Student Tasks

Q1. Identify and explain the key security threats and vulnerabilities present in the system.

Q2. What types of attacks can be performed due to lack of encryption and authentication?

Q3. How can asymmetric cryptography be applied to secure communication in this system?

Q4. Explain how digital signatures can help in verifying user identity and preventing fake requests.

Q5. What role does hashing play in ensuring data integrity in this scenario?

Q6. Suggest a secure mechanism for user authentication in the chatbot system.

Q7. How can the system ensure confidentiality, integrity, and availability (CIA triad)?

Q8. Discuss how integrating security mechanisms with AI systems improves reliability and trust.

Instructions

- Answer all questions in your own words
- Focus on information security concepts
- Provide clear and concise explanations
- Do not copy from external sources